

AudioBuddy: An Assistive Technology for People with Noise Sensitivity and Their Care Networks

Emani Hicks
dotche@uci.edu

University of California, Irvine
Irvine, California, USA

ABSTRACT

Noise sensitivity is prevalent in both neurodivergent and neurotypical people, making self-regulation challenging and impacting their quality of life. Little work has been done in the wearable and ubiquitous computing space to support these particular experiences. My research explores how people with noise sensitivity (PWNS) and those around them manage and regulate their own and other's reactions to noise. Leveraging wearable and mobile devices, this work explores the management and regulation behaviors of PWNS and their care networks. It describes the design and deployment process of a novel application that utilizes wearable sensor technology to sense physiological and environmental signals, facilitate awareness and information sharing, and present users with strategies for managing and regulating noise sensitivity experiences. This research strives to contribute an understanding of noise sensitivity experiences, inclusive design practices, and a technological solution to support people with noise sensitivity.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing.**

KEYWORDS

noise sensitivity, assistive technology, collaborative technology, wearable and sensing technology

ACM Reference Format:

Emani Hicks. 2024. AudioBuddy: An Assistive Technology for People with Noise Sensitivity and Their Care Networks. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3678365>

1 INTRODUCTION

Over half of autistic people and a large portion of non-autistic people report experiencing noise sensitivity, such as discomfort with noises in various contexts [2, 29]. Self-regulation under these conditions can be challenging, leading to responses ranging from

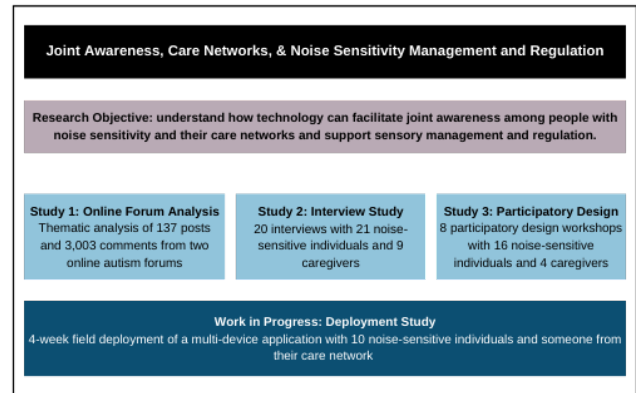


Figure 1: Dissertation Overview

fleeing an area to self-injury [11, 26, 29]. Strategies like sound masking and avoidance behaviors are suboptimal, impacting situational awareness, comfort, stigma, cost, and dependency [15]. This underscores the need for better regulation and management practices. Additionally, noise sensitivity is often managed and regulated by the person experiencing it or by someone with enough awareness and power to intervene. Yet, there are few times when the person with noise sensitivity and another come together to manage and regulate the experience. Research focused on people with noise sensitivity (PWNS) and collaborative methods for supporting their sensory needs through technological interventions is minimal.

1.1 Thesis Overview

My thesis posits that wearable and mobile technology can facilitate awareness and collaborative management and regulation of noise sensitivity experiences between PWNS and their care networks. In particular, technology that integrates physiological and environmental sensing, self and co-regulation strategies, and information sharing 1) can better help people be aware and 2) collaboratively manage and regulate noise sensitivity experiences. To investigate these claims, my prior work has informed the design and development of a technological probe called AudioBuddy, consisting of a mobile and smartwatch app to enable PWNS and someone from their care network to collaboratively manage and regulate their noise sensitivity experiences. Over the past three years, I have studied how people make sense of their noise sensitivity, the role of awareness and collaboration in management, and the challenges they experience, informing how technology can be designed to support their management and regulation needs (see Figure 1).



This work is licensed under a Creative Commons Attribution International 4.0 License.

UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1058-2/24/10.
<https://doi.org/10.1145/3675094.3678365>

1.2 Expected Contributions

Through my dissertation research, I will make several key contributions. First, I will provide a comprehensive understanding of the lived experiences of PWNS and their care networks. Building upon this foundation, I will contribute a theoretical approach to the role of wearable and mobile technology in collaborative care around invisible disabilities. Finally, I will demonstrate the ability of wearable and mobile technology to facilitate awareness and support PWNS and their care networks in better managing and regulating noise sensitivity experiences.

2 RELATED WORK

2.1 Wearable and Ubiquitous Technology

Prior research focused on neurodivergent populations and self-regulation has explored ways to use wearable devices to track personal health-related data and support their self-regulation. The most common wearable devices used were wrist-worn, such as smartwatches, either collecting physiological data to provide personalized support [4, 27, 28] or providing technological interventions without collecting any physiological data [10, 14, 24, 32]. For instance, Costa et al. designed a smartwatch-based intervention device called BoostMeUp that detects fast-beating heart rates and simulates slow-pacing artificial heart rates through haptic feedback to override the user's heart rate, thus helping the user in self-regulating their anxiety and improve their cognitive performance [4]. Simm et al. [24] also conducted workshops where autistic participants used DIY kits to create their own digital stretch wristband called Snap that records the physical interactions to provide more personalized interventions later. After four weeks of an evaluation study tracking the participants' anxiety levels, they discovered that all the participants were highly engaged and had positive experiences with using Snap for anxiety regulations [24].

Several works explored other types of wearable devices that collect physiological data for biofeedback to support regulation, such as electroencephalogram (EEG) recording devices that track brain activity [12, 21] or breathing pacer systems [3, 18, 25, 31]. For example, Park et al. designed a game platform that tracks the participants' electrical brainwaves to analyze their attention and emotional states with the purpose of helping children with ADHD improve their level of attention, regulate their hyperactivity, and develop social skills [21]. Further, Lyu et al. [17] designed Eggly, a mobile game that integrates augmented reality (AR) and neurofeedback training (NFT) to support autistic children, has shown how gamified and personalized interventions can affect engagement with interventions, especially when working with neurodiverse populations.

Research specifically for the management of noise sensitivity has explored the use of noise-attenuating headphones, hearing aids for tinnitus, hearables, and, more recently, augmented reality and virtual reality [23]. Still, work within human-computer interaction is under explored, especially in the domain of wearable and ubiquitous technology to support PWNS, with the exception of Yap et al.'s [30] work. This study focuses on App4Autism, which uses a pulse sensor and mobile application to deliver music therapy to minimize noise sensitivity responses when an increase in noise is detected for children with autism.

In summary, this prior work shows the applicability of wearable and ubiquitous systems to aid self-regulation in various contexts. With these insights, I explore the potential of wearable and ubiquitous technologies to support the noise sensitivity experiences of PWNS and their care networks.

3 METHODS

Over the past three years, I have employed a multi-method approach as part of a larger research team to investigate how PWNS and those around them manage and regulate their own and others' reactions to noise. I have explored how people make sense of their noise sensitivity, the role of awareness and collaboration, and what PWNS and caregivers consider when designing technology to support management and regulation needs.

3.1 Content Analysis of Online Forums

Content analysis of online communities has been shown to be an effective method for gathering formative information about a population of interest (see [1, 13].) Thus, to understand how PWNS experience and manage noise sensitivity, I, as part of a team, conducted a content analysis in which we analyzed interactions of autistic individuals in two online autism forums as they discussed, made sense of, and supported each other regarding their self-described noise sensitivities [8]. Two online forums described as spaces for autistic people, open for public viewing, and permission to use information from the forum were identified to be included in the study. Once I identified the forums I would use, I used the following keywords to identify posts of interest for retrieval: noise sensitivity, sound sensitivity, meditation, music, and coping. Along with my collaborators, I analyzed 12 years of posts and comments ranging from December of 2010 to February of 2022. Once posts were identified, we analyzed a preliminary subset of 36 posts and 133 comments. Following the preliminary analysis, I systematically identified an additional set of posts and comments and included a total of 137 and 3,003 comments in the final analysis.

3.2 Interview Study

In prior literature, awareness and collaboration were important for successfully managing the health and well-being of people experiencing various conditions. The data from the forum analysis contained hints of some form of awareness and collaboration but not enough to gather a clear understanding of how it functioned. My collaborators and I conducted an interview study to explore PWNS's management and regulation practices and their care networks [9]. We recruited participants ages eight and older who spoke English and self-reported sensitivity to sound. In total, 30 people participated in the study (8 males, 19 females, and three non-binary/gender fluid) between the ages of 9 and 69 (average age=38). Of the participants, ten caregiver-child pairs (20 people) and ten individuals completed one sixty-minute interview. During the interviews, participants reflected on their experiences of noise sensitivity, management styles and processes, and potential ways technology could support them.

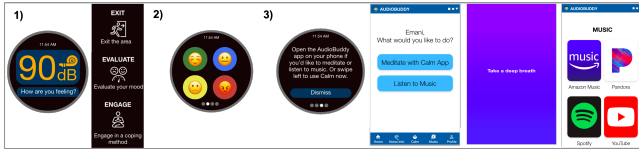


Figure 2: AudioBuddy System. 1) Noise notification with the support protocol. 2) Evaluation through the Zones of Regulation. 3) Engage in a coping strategy to support emotion regulation.

3.3 Participatory Design

My latest work used participatory design to inform the development of future technologies to support PWNS and their care networks. Eight participatory design workshops were conducted with 16 PWNS and 4 caregivers¹ (8 males, 11 females, and 1 non-binary/gender fluid) between the ages of 9 and 69 years old. Of the participants, six caregiver-child pairs (12 people) and four individuals completed the workshops (pilot workshops not included). I conducted two pilot workshops and six larger workshops, for a total of eight, consisting of activities where participants shared their experiences with noise and informed design parameters for wearable and ubiquitous technology to support noise sensitivity and collaboration through prototyping activities.

4 EVALUATION

4.1 Understanding the Experiences of PWNS

The online analysis revealed three themes: sharing experiences of noise sensitivity, managing noise sensitivity, and disclosing autism. Key findings show several societal barriers and challenges impacting noise sensitivity management and coping preferences, ranging from inaccessible environments to a lack of home, school, and workplace accommodations. Taking note of current management practices, design implications for wearable and sensor technology were provided, identifying that such technology can be used to monitor environmental and physiological climates to send feedback to users to increase awareness and provide personalized approaches for managing noise sensitivity based on certain contexts. This work informed the preliminary design of our assistive application, AudioBuddy [6] (see Figure 2). The initial version of AudioBuddy was developed for WearOS and Android devices. It included notifications of increased noise levels, supported self-regulation and emotion regulation principles from the Zones of Regulation² [16], and educational materials for self-advocacy.

4.2 Accessibility Through Awareness

The interview study suggests that to support regulation during noise sensitivity experiences, the person with noise sensitivity or someone with enough influence to respond and manage must be aware that 1) the sensitivity exists and 2) something such as a sound, person, or environment is triggering that sensitivity. This

¹Some of the participants that were noise sensitive were also caregivers, but I only counted them once as PWNS.

²The Zones of Regulation is a social emotional learning curriculum designed for self-regulation and emotion regulation that teaches how to manage and regulate their feelings. More information can be found at <https://zonesofregulation.com/>

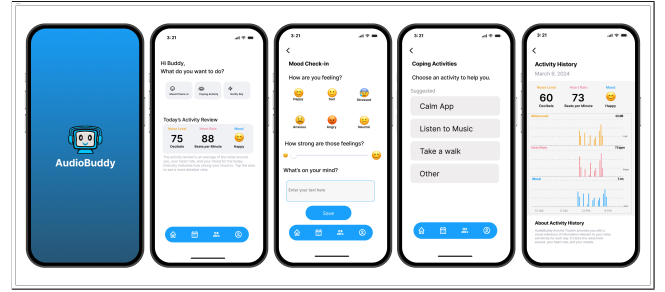


Figure 3: AudioBuddy System. 1) Splash screen, home screen, mood check-in and reflection, coping activities, and activity history showing the day's noise level, heart rate, and mood data.

awareness can be more general, in the moment, after the event, or some combination. Finally, in this work, I introduce the concept of joint awareness, which fosters collaborative management and regulation of noise sensitivity.

4.3 Co-designing with PWNS

The participatory design study emphasized the value of sensing and tracking technologies, such as applications to monitor body temperature, step count, blood pressure, and location, to increase awareness of individuals' well-being. Participants expressed wanting data tracking and visualizations to facilitate reflection and comprehension for themselves and those in their care network. Customization of tools to offer context-specific self-regulation strategies was highlighted as essential, with the design accommodating family involvement in coping strategies, which was particularly noted among parent designers. Finally, I identified inclusive practices for involving children with noise sensitivity in the design process [7], which had further implications in child-centered AI design and testing [5]. In summary, this latest study indicates a strong avenue for wearable and ubiquitous technologies, specifically regarding opportunities for sensing and tracking, personalization, and self- and co-regulation.

5 DESIGNING AUDIOBUDDY

Our earlier studies and an understanding of the literature have led to the latest redesign of AudioBuddy (Figure 3). It operates in two modes: an event-triggered scenario where the application sends a notification based on tracked physiological (*e.g.*, heart rate) and environmental data from the smartwatch, prompting the users to engage in the app, or a manual mode through independent engagement. AudioBuddy includes a toolbox of regulation strategies like breathing exercises, music listening, and physical activities, with users rating the effectiveness of each tool for a more personalized experience. The system also facilitates awareness through emotion check-ins, journal entries, and data visualizations. Finally, users can share their information with someone in their care network to increase their awareness, empower them to self-advocate, or to engage in collaborative management and regulation strategies when needed.

6 FUTURE WORK: DEPLOYMENT STUDY

For my final dissertation study, I will demonstrate how wearable and mobile technology can 1) facilitate awareness and 2) lead to collaborative management and regulation of noise sensitivity experiences between PWNS and their care networks. To do this, I will deploy AudioBuddy as a technological probe to enable PWNS and someone from their care network to manage and regulate noise sensitivity experiences. I aim to recruit PWNS who are ages 13 and older and someone from their support network (e.g., a parent, partner, sibling, or friend), hereafter identified as allies, to participate in a four-week field deployment of the application. The primary users of the application will be PWNS, with their allies positioned as peripheral users. The purpose of the field deployment is to use AudioBuddy as a probe for exploring how to facilitate awareness (e.g., self, others, and joint) among PWNS to support their management and regulation strategies better and capture how they use the tool to do so. With PWNS being the primary users, the app will allow them to receive support for managing and regulating their responses to noise through mood evaluations, coping activities, and daily reflections. It also allows them to notify and share data with their allies through either push notifications or in-app notifications, thus enabling opportunities for their allies to become aware and support as needed. Allies that are recruited to participate will also use the AudioBuddy system in the same way in which the noise sensitive user does to understand the experiences of the PWNS, especially when environments are loud [9]. However, their primary usage will be to receive notifications about sensory experiences and data shared with them by PWNS to enhance their own awareness and engage in co-regulation and collaborative management of the situation when needed.

6.1 In-lab Usability Evaluation and Pilot Deployment

Before deploying, I will conduct a usability evaluation and an in-lab deployment study with members from my lab and within my department. These evaluations will assess the app's user interface and usability and identify areas for improvement in the system's design. For the usability evaluation, I will use heuristics [20] and the think-aloud approach [19] as methods for assessment. A month before the full deployment, I will conduct the in-lab pilot deployment, during which the evaluators will use the app for a week and engage in all the field deployment activities. The data from these evaluations will be analyzed, and based on this analysis, the app will be revised.

6.2 Field Deployment

At the start of the deployment study, participants will undergo an onboarding interview where they will be provided with an Apple Watch and iPhone with the AudioBuddy system on it. This interview will include a walk-through of the app and an opportunity to ask questions about the process. After each week of engagement, I will conduct interviews. Interviews will consist of questions related to self-awareness and regulation, data-sharing practices with their allies to enhance their awareness and their perceptions about information sharing. During the four-week period, participants may

receive notifications about environmental or physiological indicators of potential noise sensitivity incidents when specific thresholds are triggered, prompting them to engage in a regulation activity and complete an in-app survey to document these experiences. Additionally, participants will receive daily reminders encouraging them to complete mood evaluations to support emotion regulation [7, 22]. At the end of each day, participants will also complete a web-based survey to reflect on their overall experiences. After engaging with the application, I will conduct a focus group with both the PWNS and their allies to explore how we can further improve the system for cooperative support. Finally, quantitative analyses will be conducted using passive sensed data (e.g., heart rate, ambient sound level, accelerometer, etc.) to identify correlations between passive sensed data and self-reported data.

7 EXPECTED CONTRIBUTIONS

Through this project, I will demonstrate the potential of wearable and mobile technology to facilitate awareness and collaboration between PWNS and their allies for noise sensitivity management and regulation. By exploring correlations in passive sensing data and self-reported data, this work offers insights that may interest ubiquitous computing scholars by providing an avenue for possible applications for machine learning and artificial technology. Finally, in exploring the perception and effect of information sharing on privacy and security concerns between PWNS and their allies, potential contributions may include principles for data-sharing practices and privacy by design in social computing.

ACKNOWLEDGMENTS

I thank Gillian Hayes, Aaron Adams, Jazette Johnson, Elizabeth Ankrah, Lucas Silva, Aehong Min, Franceli Cibrian, LouAnne Boyd, Arpita Bhattacharya, Weijie (Chris) Du, Jialuo (Jerry) Hu, Avery Mavrovounioti, Kade Na, Nathan Serrano, Rafael Carrillo Munoz, Sara Chong, Jessica Choi, Hanna Mofid, and Jessica Sotelo for their support throughout my research journey and the research participants for their collaboration and input. I also thank Armando Beltrán, Arturo Morales Téllez, Ashhad Shah, Kohsuke Hirano, and Mahan Tafreshipour for their support in developing the AudioBuddy system. This research is supported by the National Science Foundation Graduate Fellowship under Grant No. DGE-1839285 and support from the Jacobs Foundation CERES Network.

REFERENCES

- [1] Anietie Andy and Uduak Andy. 2021. Understanding Communication in an Online Cancer Forum: Content Analysis Study. *JMIR Cancer* 7, 3 (7 Sep 2021), e29555. <https://doi.org/10.2196/29555>
- [2] Elaine N Aron. 2013. *The highly sensitive person: How to thrive when the world overwhelms you*. Kensington Publishing Corp.
- [3] R Bossenbroek, J Weerdmeester, A Granic, M van Rooij, et al. 2020. Efficacy of a virtual reality biofeedback game (DEEP) to reduce anxiety and disruptive classroom behavior: Single-case study, *JMIR Mental Health*, 7 (3), e16066. <https://doi.org/10.2196/29555>
- [4] Jean Costa, François Guimbretière, Malte F Jung, and Tanzeem Choudhury. 2019. Boostmeup: Improving cognitive performance in the moment by unobtrusively regulating emotions with a smartwatch. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 2 (2019), 1–23.
- [5] Emani Hicks and Vitica Arnold. 2024. Inclusive Practices for Child-Centered AI Design and Testing. arXiv:2404.05920 [cs]. <http://arxiv.org/abs/2404.05920>
- [6] Emani Hicks, Jesus Armando Beltran, Jazette Johnson, Ashhad Shah, Kohsuke T. Hirano, Franceli L. Cibrian, and Gillian Hayes. 2023. AudioBuddy: Using Sound

- Sensors to Support Sound Sensitivity Awareness in Autistic Individuals. In *Adjunct Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing and the 2022 ACM International Symposium on Wearable Computers* (New York, NY, USA, 2023-04-24) (*UbiComp/ISWC '22 Adjunct*). Association for Computing Machinery, 27–29. <https://doi.org/10.1145/3544793.3560336>
- [7] Emani Hicks, Jialuo Hu, Avery Mavrovounioti, Weijie Du, Jazette Johnson, Elizabeth A. Ankrah, Aehong Min, and Gillian R. Hayes. 2023. Supporting Noise Sensitivity and Emotion Regulation with Children. In *Interaction Design and Children* (2023-06-19). 7. <https://doi.org/10.1145/3585088.3593887>
- [8] Emani Hicks, Jazette Johnson, Rebecca W. Black, and Gillian R. Hayes. 2023. Understanding Noise Sensitivity through Interactions in Two Online Autism Forums. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility* (New York, NY, USA) (*ASSETS '23*). Association for Computing Machinery, New York, NY, USA, Article 19, 12 pages. <https://doi.org/10.1145/3597638.3608413>
- [9] Emani Hicks, Avery Mavrovounioti, Weijie Du, Elizabeth Ankrah, Jazette Johnson, Aehong Min, and Gillian R. Hayes. 2024. Accessibility through Awareness of Noise Sensitivity Management and Regulation Practices. In *The 26th International ACM SIGACCESS Conference on Computers and Accessibility* (St. John's, NL, Canada) (*ASSETS '24*). Association for Computing Machinery, New York, NY, USA, 18 pages. <https://doi.org/10.1145/3663548.3675630>
- [10] Anya S Evmenova, Heidi J Graff, Vivian Genaro Motti, Kudirat Giwa-Lawal, and Hui Zheng. 2019. Designing a wearable technology intervention to support young adults with intellectual and developmental disabilities in inclusive postsecondary academic environments. *Journal of Special Education Technology* 34, 2 (2019), 92–105.
- [11] Pawel J. Jastreboff and Margaret M. Jastreboff. 2014. Treatments for Decreased Sound Tolerance (Hyperacusis and Misophonia). 35, 2 (2014), 105–120. <https://doi.org/10.1055/s-0034-1372527> Publisher: Thieme Medical Publishers.
- [12] Han Jiang and Stuart J Johnstone. 2015. A preliminary multiple case report of neurocognitive training for children with AD/HD in China. *SAGE Open* 5, 2 (2015), 2158244015586811.
- [13] Jazette Johnson, Rebecca W. Black, and Gillian R. Hayes. [n. d.]. Roles in the Discussion: An Analysis of Social Support in an Online Forum for People with Dementia. 4 ([n. d.]). Issue CSCW2. <https://doi.org/10.1145/3415198> Place: New York, NY, USA Publisher: Association for Computing Machinery.
- [14] Niloofer Kalantari, Hui Zheng, Heidi J. Graff, Anna S. Evmenova, and Vivian Genaro Motti. 2021. Emotion Regulation for Neurodiversity through Wearable Technology. In *2021 9th International Conference on Affective Computing and Intelligent Interaction (ACII)*. 1–8. <https://doi.org/10.1109/ACII52823.2021.9597458>
- [15] Ankit Koirala, Zhiwei Yu, Hillary Schiltz, Amy Van Hecke, Kathleen A Koth, and Zhi Zheng. 2019. An Exploration of Using Virtual Reality to Assess the Sensory Abnormalities in Children with Autism Spectrum Disorder. In *Proceedings of the 18th ACM International Conference on Interaction Design and Children* (New York, NY, USA, 2019-06-12) (*IDC '19*). Association for Computing Machinery, 293–300. <https://doi.org/10.1145/3311927.3323118>
- [16] Leah Kuypers. 2011. The zones of regulation. *San Jose: Think Social Publishing* (2011).
- [17] Yue Lyu, Pengcheng An, Yage Xiao, Zibo Zhang, Huan Zhang, Keiko Katsuragawa, and Jian Zhao. 2023. Eggly: Designing Mobile Augmented Reality Neurofeedback Training Games for Children with Autism Spectrum Disorder. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 7, 2, Article 67 (jun 2023), 29 pages. <https://doi.org/10.1145/3596251>
- [18] Pardis Miri, Mehul Arora, Aman Malhotra, Robert Flory, Stephanie Hu, Ashley Lowber, Ishan Goyal, Jacqueline Nguyen, John P Hegarty, Marlo D Kohn, et al. 2022. FAR: End-to-End Vibrotactile Distributed System Designed to Facilitate Affect Regulation in Children Diagnosed with Autism Spectrum Disorder Through Slow Breathing. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–20.
- [19] Janni Nielsen, Torkil Clemmensen, and Carsten Yssing. 2002. Getting access to what goes on in people's heads? reflections on the think-aloud technique. In *Proceedings of the Second Nordic Conference on Human-Computer Interaction* (Aarhus, Denmark) (*NordiCHI '02*). Association for Computing Machinery, New York, NY, USA, 101–110. <https://doi.org/10.1145/572020.572033>
- [20] Jakob Nielsen and Rolf Molich. 1990. Heuristic evaluation of user interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Seattle, Washington, USA) (*CHI '90*). Association for Computing Machinery, New York, NY, USA, 249–256. <https://doi.org/10.1145/97243.97281>
- [21] Kyungeun Park, Taesuk Kihl, Seungie Park, Min-Jae Kim, and Juno Chang. 2016. Narratives and sensor driven cognitive behavior training game platform. In *2016 IEEE 14th International Conference on Software Engineering Research, Management and Applications (SERA)*. IEEE, 125–131.
- [22] L. J. Rinaldi, J. Simner, S. Koursarou, and J. Ward. [n. d.]. Autistic traits, emotion regulation, and sensory sensitivities in children and adults with Misophonia. ([n. d.]). <https://doi.org/10.1007/s10803-022-05623-x>
- [23] Grant D. Searchfield, Philip J. Sanders, Zohreh Doborjeh, Maryam Doborjeh, Roger Boldu, Kevin Sun, and Amit Barde. 2021. A State-of-Art Review of Digital Technologies for the Next Generation of Tinnitus Therapeutics. *Frontiers in Digital Health* 3 (2021). <https://doi.org/10.3389/fdgh.2021.724370>
- [24] Will Simm, Maria Angela Ferrario, Adrian Gradinar, Marcia Tavares Smith, Stephen Forshaw, Ian Smith, and Jon Whittle. 2016. Anxiety and autism: towards personalized digital health. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. 1270–1281.
- [25] Tobias Sonne and Mads Møller Jensen. 2016. Chillfish: A respiration game for children with adhd. In *Proceedings of the TEI'16: Tenth International Conference on Tangible, Embedded, and Embodied Interaction*. 271–278.
- [26] Lillian N. Stiegler and Rebecca Davis. [n. d.]. Understanding Sound Sensitivity in Individuals with Autism Spectrum Disorders. 25, 2 ([n. d.]), 67–75. <https://doi.org/10.1177/1088357610364530> Publisher: SAGE Publications Inc.
- [27] Juan C Torrado, Javier Gomez, and Germán Montoro. 2017. Emotional self-regulation of individuals with autism spectrum disorders: smartwatches for monitoring and interaction. *Sensors* 17, 6 (2017), 1359.
- [28] Juan C Torrado, Javier Gomez, and Germán Montoro. 2019. Assisted behavior control with wearable devices: A case study for high-functioning autism. *Multidisciplinary Digital Publishing Institute Proceedings* 31, 1 (2019), 2.
- [29] Zachary J. Williams, Jason L. He, Carissa J. Cascio, and Tiffany G. Woynaroski. 2021. A review of decreased sound tolerance in autism: Definitions, phenomenology, and potential mechanisms. 121 (2021), 1–17. <https://doi.org/10.1016/j.neubiorev.2020.11.030>
- [30] Cheng Yee Yap, Kher Hui Ng, Yungen Cheah, Suk Yee Lim, Jessica Price, and Marieke De Vries. [n. d.]. App4Autism: An Integrated Assistive Technology with Heart Rate Monitoring for Children with Autism. In *Advances in Visual Informatics* (Cham, 2019), Halimah Badioze Zaman, Alan F. Smeaton, Timothy K. Shih, Sergio Velastin, Tada Terutoshi, Nazlena Mohamad Ali, and Mohammad Nazir Ahmad (Eds.). Springer International Publishing, 498–512. https://doi.org/10.1007/978-3-030-34032-2_44
- [31] M Abdullah Zafar, Beena Ahmed, Rami Al Rihawi, and Ricardo Gutierrez-Osuna. 2018. Gaming away stress: Using biofeedback games to learn paced breathing. *IEEE Transactions on Affective Computing* 11, 3 (2018), 519–531.
- [32] Hui Zheng and Vivian Genaro Motti. 2018. Assisting students with intellectual and developmental disabilities in inclusive education with smartwatches. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. 1–12.